A Comparative Study of Machine Learning Models for Loan Recovery and Default Prediction

Machine Learning

Submitted by

Prateek Gupta

S243DEC28

MA Economics

Dr. B.R. Ambedkar University Delhi

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1. INTRODUCTION

Loan default prediction is a critical task in credit risk management, as it enables financial institutions to assess the likelihood of loan recovery and minimize potential financial losses. With the growing volume and complexity of lending data, machine learning techniques have become effective tools for improving the accuracy and reliability of default prediction models.

This project aims to develop a robust machine learning framework capable of accurately predicting loan defaults using historical lending data. The objective is to equip financial institutions with a data-driven decision-support tool that can help mitigate credit risk, optimize lending strategies, and improve operational efficiency.

The scope of the project includes several key stages: data loading and initial exploration to understand the structure of the dataset; exploratory data analysis (EDA) to identify patterns, relationships, and anomalies; data preprocessing involving data cleaning, handling missing values, encoding categorical variables, and feature engineering; and model training using multiple classification algorithms such as Decision Tree, Random Forest, XGBoost, and CatBoost. The trained models are evaluated and compared using standard performance metrics including Accuracy, AUC, and F1-score to identify the most effective approach for loan default prediction.

Overall, the project seeks to provide actionable insights that can enhance credit risk assessment and support more informed decision-making in lending institutions.

2. DATASET DESCRIPTION

The dataset used in this study originates from LendingClub, a peer-to-peer lending platform, and contains detailed loan-level information covering the period from 2007 to the fourth quarter of 2018. The dataset includes comprehensive records related to loan applications, borrower demographics, credit characteristics, loan performance, and other relevant financial indicators, making it well suited for loan default and recovery analysis.

Initially, the dataset consisted of 2,260,701 observations with 151 variables. To ensure meaningful analysis, the data was filtered to include only completed loan statuses, namely Fully Paid, Charged Off, Default, and late-stage loans. After this filtering process, the final dataset was reduced to 1,373,154 observations while retaining all 151 variables.

The dataset contains both numerical and categorical features capturing borrower financial health, credit behavior, and loan attributes. Key variables used in this study are described below :-

Table 1: Key Variables and Descriptions

Variable Name	**Description**	**Status**
loan_amnt	Loan amount requested by the borrower	Available
int_rate	Interest rate charged on the loan	Available
installment	Monthly loan installment amount	Not Found
annual_inc	Annual income of the borrower	Available
dti	Debt-to-income ratio	Available
fico_score	Borrower's FICO credit score	Not Found
revol_bal	Total revolving balance	Available
revol_util	Revolving credit utilization percentage	Available
out_prncp	Outstanding principal remaining	Available
last_pymnt_amnt	Amount of the most recent payment	Available
tot_cur_bal	Total current balance across all accounts	Available
total_rev_hi_lim	Total revolving credit limit	Available
emp_length	Employment length in years	Available
home_ownership	Home ownership status of the borrower	Available
purpose	Purpose of the loan	Available

Together, these variables provide a comprehensive representation of borrower risk profiles and loan performance, enabling effective modeling and evaluation of loan default prediction using machine learning techniques.

3. TARGET VARIABLE AND LOSS FUNCTIONS

3.1 Target Variable Definition

The target variable for this binary classification problem is *loan_status_binary*. This variable was derived from the original *loan_status* column by mapping multiple loan statuses into two distinct outcome classes. Loans that were successfully repaid were assigned a value of 0 (Paid), while loans that defaulted or exhibited late repayment behavior were assigned a value of 1 (Default). This transformation simplifies the prediction task and enables effective modeling of loan default risk.

Specifically, the mapping was performed as follows:

- **Paid (0):** *Fully Paid*, *Does not meet the credit policy. Status: Fully Paid*
- **Default (1):** *Charged Off*, *Default*, *Late (31–120 days)*, *Late (16–30 days)*, *Does not meet the credit policy. Status: Charged Off*

3.2 Class Distribution

The dataset exhibits an imbalanced distribution of the target variable *loan_status_binary*. Out of the total observations, 1,078,739 loans (78.6%) belong to the Paid class, while 294,415 loans (21.4%) are classified as Default. This imbalance indicates that default cases form a smaller but economically significant portion of the dataset. Such imbalance can influence model performance and necessitates careful evaluation using appropriate performance metrics beyond overall accuracy.

3.3 Loss Functions and Model Objectives

The machine learning models employed in this study utilize loss functions and optimization criteria suitable for binary classification tasks.

For gradient boosting models, the *XGBClassifier* was explicitly configured with the logarithmic loss function (*logloss*), also known as binary cross-entropy. This loss function evaluates the difference between predicted probabilities and actual class labels, penalizing confident but incorrect predictions more heavily.

Similarly, the *CatBoostClassifier* uses a variant of logarithmic loss as its default objective for binary classification. The model optimizes this loss function to improve probabilistic predictions while effectively handling categorical features and reducing overfitting.

In contrast, the *RandomForestClassifier* and *DecisionTreeClassifier* do not directly minimize a probabilistic loss function. Instead, they rely on impurity-based splitting criteria such as Gini impurity or entropy during tree construction. These criteria aim to create homogeneous child nodes by reducing classification uncertainty at each split, thereby improving predictive performance. Although their optimization approach differs from gradient boosting models, the resulting predictions can still be interpreted in a probabilistic framework for classification tasks.

4. DATA PREPROCESSING

This section summarizes the data preprocessing steps performed prior to model training to ensure data consistency, quality, and suitability for machine learning algorithms.

4.1 Cleaning of String-Based Features

Several features were originally stored as strings and were converted into numerical formats. The interest rate (*int_rate*) and revolving utilization (*revol_util*) variables were transformed by removing the percentage symbol and converting the values to floating-point numbers. Employment length (*emp_length*) was processed by extracting the numerical representation of years from textual descriptions, with missing values initially assigned a value of zero. The loan term (*term*) was converted from string format to numerical values representing the number of months.

4.2 Creation of Unified FICO Score

A unified credit score variable, *fico_score*, was constructed by taking the average of *fico_range_low* and *fico_range_high*. This transformation provided a single, continuous measure of borrower creditworthiness for modeling purposes.

4.3 Feature Selection

A subset of relevant features was selected based on their importance in capturing borrower characteristics, credit behavior, and loan performance. The selected numerical features include *loan_amnt* (loan amount requested), *int_rate* (interest rate charged on the loan), *annual_inc* (annual income of the borrower), *dti* (debt-to-income ratio), *fico_score* (average of FICO score range), *revol_bal* (total revolving credit balance), *revol_util* (revolving credit utilization percentage), *out_prncp* (outstanding principal balance), *last_pymnt_amnt* (amount of the most recent payment), *tot_cur_bal* (total current balance across all accounts), *total_rev_hi_lim* (total revolving credit limit), and *emp_length* (length of employment in years).

In addition, categorical features capturing borrower and loan characteristics were included, namely *home_ownership* (borrower's home ownership status) and *purpose* (primary purpose of the loan). Feature availability was verified prior to model training, and variables not directly present in the dataset were either derived during preprocessing or excluded from the final feature set.

4.4 Handling Missing Values

Missing values in numerical features were imputed using the median of the respective columns to reduce the influence of outliers. For categorical variables, missing values were imputed using the mode, ensuring that the most frequently occurring category was assigned.

4.5 Encoding of Categorical Variables

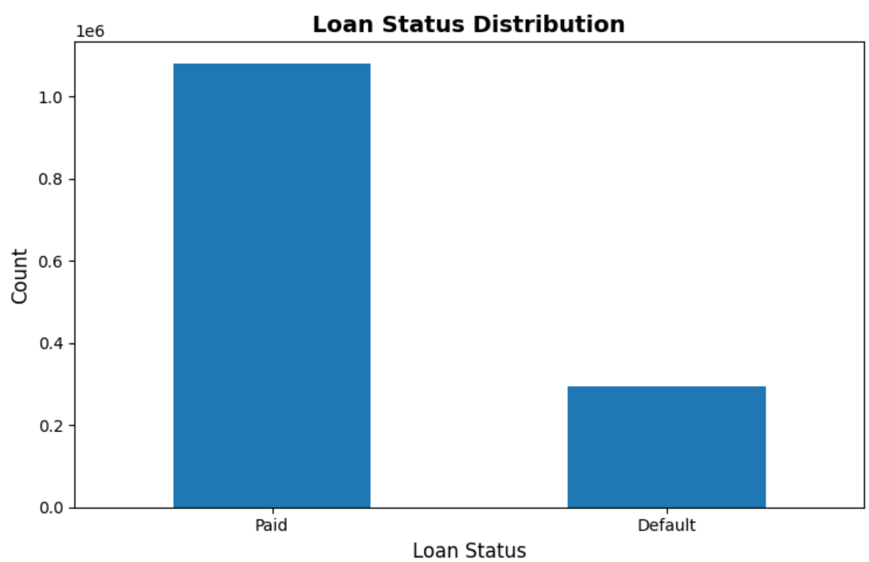
Categorical features were transformed into numerical representations using label encoding. The variables *home_ownership* and *purpose* were encoded into integer-based categories, resulting in new features suitable for model input.

4.6 Final Feature Set and Data Splitting

The final feature set used for model training consisted of all cleaned numerical variables and the encoded categorical features. The dataset was then divided into training and testing sets using a 70:30 split. A random state of 42 was applied to ensure reproducibility. The distribution of the target variable, *loan_status_binary*, was verified prior to splitting and maintained approximately 78.6% Paid (0) and 21.4% Default (1) across both subsets.

5. EXPLORATORY DATA ANALYSIS

This section summarizes the key insights obtained from the Exploratory Data Analysis (EDA) phase, which was conducted to understand the structure, distribution, and relationships within the dataset prior to model development.



5.1 Initial Filtering and Data Overview

The original dataset consisted of over 2.2 million observations with 151 variables. To ensure analytical relevance, the data was filtered to include only completed loan records. This filtering process resulted in a refined dataset comprising approximately 1.37 million observations while retaining all original variables.

5.2 Target Variable Analysis

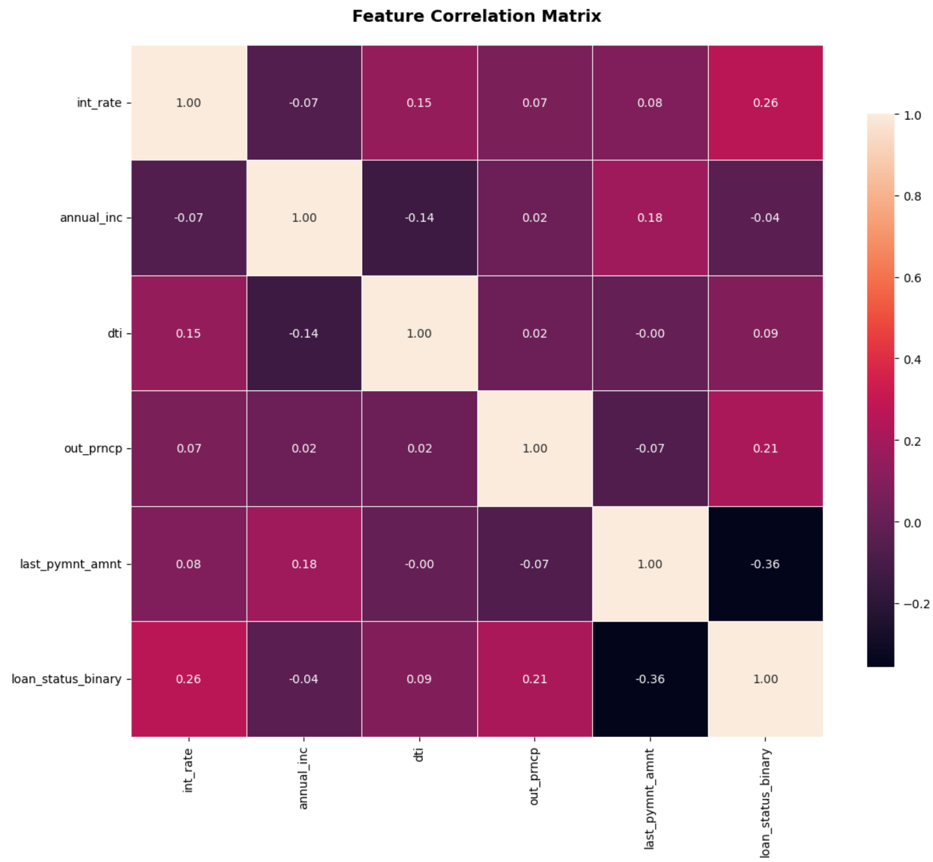
The target variable, *loan_status_binary*, was constructed to classify loans into two categories: Paid (0) and Default (1). An examination of the class distribution revealed a significant imbalance, with approximately 78.6% of loans classified as Paid and 21.4% classified as Default. This imbalance was visualized using bar and pie charts and highlights the importance of carefully evaluating model performance, as accuracy alone may be misleading.

5.3 Distribution of Numerical Features

The distributions of key numerical variables were examined using histograms. The interest rate (*int_rate*) appeared approximately normally distributed with a slight right skew. Annual income (*annual_inc*) exhibited a heavily right-skewed distribution, indicating that most borrowers fall within lower income ranges, with a small number of high-income outliers. The debt-to-income ratio (*dti*) was also right-skewed, showing a concentration at lower values but with a long tail. The outstanding principal (*out_prncp*) displayed a multi-modal distribution, likely reflecting variations in loan status and repayment stages. The last payment amount (*last_pymnt_amnt*) was highly right-skewed, with most payments being relatively small and a few large payments corresponding to full loan payoffs.

5.4 Relationship Between Numerical Features and Loan Status

Box plots were used to examine the relationship between numerical features and the target variable. Loans that defaulted generally exhibited higher median interest rates and greater variability compared to fully paid loans, suggesting a positive association between interest rates and default risk. Defaulted loans also showed slightly lower median annual income and higher median debt-to-income ratios, indicating that borrowers with weaker financial capacity are more prone to default. As expected, loans classified as Default showed higher outstanding principal balances, while fully paid loans had significantly higher last payment amounts, reflecting loan closure through final repayments.



5.5 Feature Correlation Analysis

A correlation heatmap was generated to quantify relationships between numerical variables and the target variable. The target variable *loan_status_binary* showed a moderate positive correlation with interest rate (0.26) and outstanding principal (0.21), reinforcing earlier observations. A moderate negative correlation was observed with last payment amount (-0.36), indicating that higher final payments are associated with loan repayment. Debt-to-income ratio showed a weak positive correlation (0.08), while annual income had a very weak negative correlation (-0.04). Inter-feature correlations were generally low to moderate, suggesting that the selected variables capture distinct aspects of borrower behavior and financial risk.

6. MACHINE LEARNING MODELS IMPLEMENTED

This project involved the implementation and training of several supervised machine learning models for a binary classification task aimed at predicting loan defaults. The selected models represent both traditional tree-based methods and advanced ensemble learning techniques, enabling a comprehensive comparison of predictive performance and computational efficiency.

The following machine learning models were implemented and evaluated in this study:

- Decision Tree Classifier
- Random Forest Classifier
- XGBoost Classifier
- CatBoost Classifier

6.1 Evaluation Metrics

To evaluate the performance of the machine learning models for loan default prediction, multiple metrics were used to capture different aspects of classification quality, particularly given the imbalanced nature of the dataset.

6.1.1 Accuracy

Accuracy measures the proportion of correctly classified Paid and Default loans out of the total observations. Although easy to interpret, accuracy can be misleading in imbalanced datasets, as high values may be achieved by favoring the majority class while failing to detect default cases.

6.1.2 Area Under the ROC Curve (AUC)

AUC measures the model's ability to distinguish between Paid and Default loans across all classification thresholds. Values range from 0.5 (random performance) to 1.0 (perfect discrimination). AUC is especially suitable for imbalanced datasets, as it evaluates class separation independently of class proportions.

6.1.3 F1-Score

The F1-Score is the harmonic mean of Precision and Recall and provides a balanced measure of performance for the Default class. It is particularly important in loan default prediction, where both identifying actual defaults and minimizing false alarms are critical.

6.1.4 Cross-Validation AUC

Cross-validation was employed to assess model robustness and generalization. The mean AUC across folds provides a reliable estimate of performance on unseen data, while the standard deviation indicates stability. Lower variation across folds suggests better generalization and consistency.

7. MODEL PERFORMANCE RESULTS

7.1 Summary of Model Performance

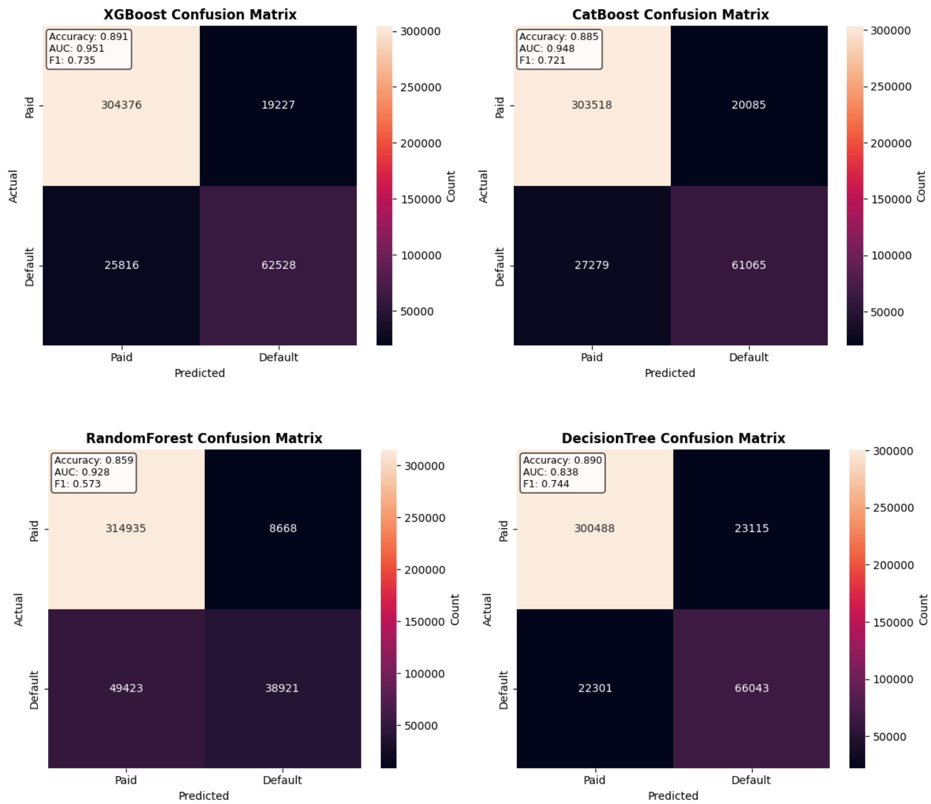
Table 2: Performance Comparison of Machine Learning Models

Model	**Accuracy**	**AUC**	**F1-Score**	**CV AUC Mean**	**CV AUC Std**
XGBoost	0.8907	0.9514	0.7352	0.9519	0.0005
CatBoost	0.8850	0.9478	0.7206	0.9477	0.0006
Random Forest	0.8590	0.9281	0.5726	0.9288	0.0006
Decision Tree	0.8898	0.8381	0.7441	0.8297	0.0025

7.2 Model-Specific Interpretation

7.2.1 Decision Tree

The Decision Tree model achieved an accuracy of 0.8898 and an AUC of 0.8381, while producing the highest F1-Score (0.7441) among all models. Its cross-validation



AUC was 0.8297 with a standard deviation of 0.0025. The confusion matrix analysis showed that the model correctly identified 66,043 defaulted loans (True Positives) while generating 23,115 False Positives and missing 22,301 actual defaults (False Negatives). The high F1-Score indicates a strong balance between precision and recall for the minority Default class, making the Decision Tree effective in identifying default risk despite its relatively lower AUC.

7.2.2 XGBoost

XGBoost demonstrated strong predictive performance with the highest AUC of 0.9514 and an accuracy of 0.8907. The F1-Score for the Default class was 0.7352, and cross-validation results showed excellent stability with a mean AUC of 0.9519 and a standard deviation of 0.0005. The model identified 62,528 True Positives, produced 19,227 False Positives, and missed 25,816 defaults. Although it identified slightly fewer defaults than the Decision Tree, XGBoost significantly reduced false positives and exhibited superior discrimination across classification thresholds.

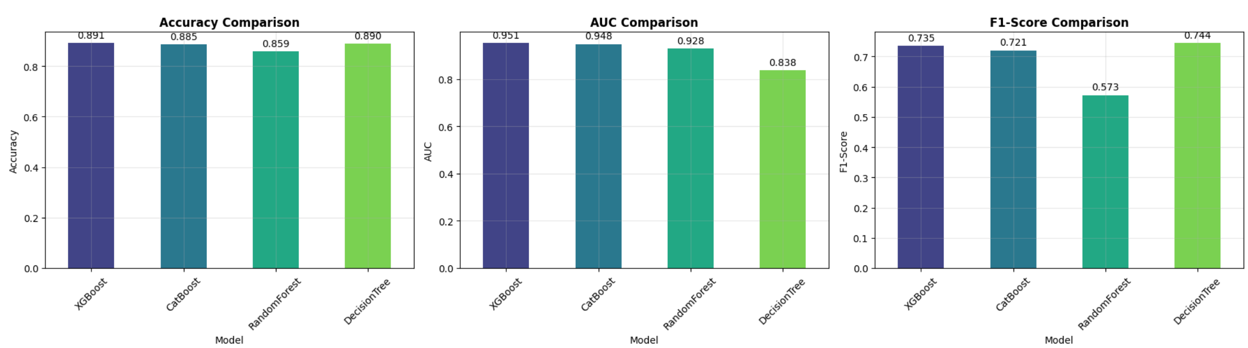
7.2.3 CatBoost

CatBoost achieved an accuracy of 0.8850 and an AUC of 0.9478, closely matching the performance of XGBoost. Its F1-Score was 0.7206, with a cross-validation AUC of 0.9477 and a standard deviation of 0.0006. The model correctly predicted 61,065 defaults, generated 20,085 False Positives, and missed 27,279 actual defaults. Overall, CatBoost showed stable and balanced performance, though with slightly lower recall for the Default class compared to XGBoost.

7.2.4 Random Forest

The Random Forest model achieved an accuracy of 0.8590 and an AUC of 0.9281 but recorded the lowest F1-Score (0.5726). Its cross-validation AUC was 0.9288 with a standard deviation of 0.0006. The confusion matrix revealed 38,921 True Positives and only 8,668 False Positives, but a high number of False Negatives (49,423). This indicates that while the model is highly precise when predicting defaults, it fails to identify a substantial proportion of actual default cases, a common outcome in imbalanced datasets when recall is not explicitly optimized.

7.3 Comparative Analysis and Trade-offs



Among the evaluated models, XGBoost exhibited the best overall discriminative ability, as reflected by the highest AUC, with CatBoost performing similarly.

The Decision Tree achieved the highest F1-Score, indicating the best balance between precision and recall for the Default class. Random Forest minimized false positives but at the cost of missing many actual defaults.

In the context of loan default prediction, the cost of false negatives is typically higher than that of false positives, as failing to identify a default can lead to significant financial losses. Therefore, models such as Decision Tree and XGBoost are more suitable for risk-sensitive applications. While the Decision Tree offers strong performance in identifying defaults, XGBoost provides superior overall discrimination and stability, making it a robust choice for real-world deployment depending on institutional risk preferences.

8. CONCLUSION

This study developed and evaluated a machine learning pipeline for loan default prediction using a large LendingClub dataset. The pipeline covered data preprocessing, exploratory analysis, model training, and performance evaluation, with the objective of supporting effective credit risk assessment.

8.1 Key Findings

- **Model Performance:** Gradient boosting models demonstrated superior overall performance. XGBoost achieved the highest AUC (0.9514), followed closely by CatBoost (0.9478), indicating strong discriminatory ability between paid and defaulted loans.

- **Default Class Identification:** The Decision Tree model achieved the highest F1-Score (0.7441), reflecting the best balance between precision and recall for the minority Default class, despite having a lower AUC (0.8381).

- **Random Forest Behavior:** Random Forest achieved a relatively high AUC (0.9281) but the lowest F1-Score (0.5726), primarily due to a high number of false negatives, indicating conservative default prediction.

8.2 Evaluation Insights

- Accuracy alone was insufficient due to class imbalance and could misrepresent true model performance.
- AUC proved effective for measuring overall class separation, while F1-Score was critical for evaluating Default class performance.
- Cross-validation AUC confirmed the stability and generalization ability of the boosting models.

8.3 Business Interpretation

In loan default prediction, false negatives typically carry higher financial risk than false positives. Considering this trade-off, XGBoost provided the most balanced performance through strong overall discrimination and reliable Default class detection, while the Decision Tree offered competitive performance when prioritizing recall for defaults.

9. REFERENCES

- LendingClub Loan Data (2007–2018). Kaggle Dataset. Available at: `https://www.kaggle.com/datasets/wordsforthewise/lending-club`